

Introduction to (Statistical) Machine Learning

Course Content and Philosophy

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Overview

1. What is Machine Learning?
2. The Theme of This Course
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Part 1: What is Machine Learning?

Machine Learning Has Several Aspects

- Machine learning is not one single subject.
- It has several intertwined aspects:
 - **Modelling**: how do we represent the data-generating process?
 - **Methods**: what algorithms do we use for learning?
 - **Computation**: how do we carry out learning efficiently?
 - **Inference**: how do we quantify uncertainty and make decisions?
- One can teach machine learning from any of these viewpoints.
- Each viewpoint can easily become its own course.

A Myriad of Models and Methods

Supervised Learning

- Linear Regression
- Logistic Regression and GLMs
- KNN
- Support Vector Machines
- Trees and Random Forests
- Neural Networks

Unsupervised / Generative Learning

- K-Means
- EM Algorithm
- Variational Autoencoders
- Diffusion Models
- GANs
- Transformers

Different methods are useful for different problems.

No One Method Dominates Everything

- There is no universally best machine learning method.
- The right method depends on the data, the scientific question being asked, the desired prediction task, the available computational resources, the need for interpretability, and the form of uncertainty one wishes to quantify.
- Over the last decade, neural networks have emerged as an extremely flexible class of function approximators.
- They have found applications across vision, language, science, engineering, and generative AI.

But neural networks do not replace the need for principles.

Part 2: The Theme of This Course

The Lens of Inference

- In this course, we will explore several machine learning methods.
- The organizing theme will be **inference**.
- More specifically, we will focus on **probabilistic machine learning**.
- Many machine learning methods arise naturally from a probability model.

Probability Model + Data $\implies$ Inference

Bayesian Statistics as a Guiding Principle

- Using probability for inference is not new.
- It is essentially the viewpoint of Bayesian statistics.
- Bayesian statistics is classical, but also deeply modern.
- It gives one core principle that can be applied to almost any learning problem.

Prior + Likelihood $\implies$ Posterior

Learning = Updating uncertainty after seeing data.

Bayesian and Frequentist Perspectives

Bayesian View

- Unknown quantities are modeled probabilistically.
- Probability represents uncertainty.
- Inference proceeds through posterior distributions.
- Prediction and uncertainty are unified.

Frequentist View

- Parameters are fixed.
- Randomness comes from repeated sampling.
- Focus on (somewhat ad hoc) estimators, tests, prediction error, and guarantees for repeated experimentation.
- Some ML tools are only justified this way.

We will emphasize Bayesian ideas, but also discuss important frequentist tools.

Part 3: Function Approximation

Function Approximation: Another Central Theme

- One of the central themes of machine learning is **function approximation**.
- In supervised learning, we observe

$$(X_1, Y_1), \dots, (X_n, Y_n)$$

and wish to understand the relationship between X and Y .

- The goal is often to construct a function

$$\hat{f}(x)$$

that predicts Y from X .

- Many machine learning methods can be viewed as different ways of estimating an unknown function.

Examples of Probabilistic Models

- A common starting point is a probabilistic model.

- Regression:

$$Y = f(X) + \varepsilon.$$

- Logistic Regression:

$$P(Y = 1 \mid X = x) = \sigma(f(x)).$$

- In each case, the key object is an unknown function.

Learning = estimating a function from data.

Different Function Classes

- Different methods correspond to different choices of function classes.
- Linear Functions and Basis expansions
 - Linear Regression
 - Logistic Regression
 - Ridge Regression
 - Lasso
 - Smoothing Splines
- Piecewise functions
 - Trend Filtering
 - Piecewise Polynomial Models
- Shape constrained functions
 - Isotonic Regression

More Function Classes

- Trees

Trees can be viewed as hierarchical piecewise constant functions.

- Neural Networks

Neural networks form a highly flexible and expressive class of functions.

- Modern machine learning often uses extremely large neural networks as generic function approximators.
- VAEs, Diffusion Models, and Large Language Models are all examples of this paradigm.

A Unifying View of Machine Learning

Many machine learning methods can be viewed as arising from three ingredients:

Probabilistic Model



Function Class



Optimization / Inference Procedure

Variational AutoEncoders, Diffusion Denoising Probabilistic Models and Transformers are complicated models of probability distributions on large spaces parametrized by functions of several/many variables.

Some Exceptions

- Not every machine learning method fits neatly into this framework.
- Local Averaging methods such as
 - k -Nearest Neighbors,
 - Local Polynomial Regression,
 - Kernel Smoothingdo not start with a function class.
- We will start with our first function class: linear functions.

Part 3: Course Content

Broad Structure of the Course

- The course will be broadly divided into two halves.
- **First Half:**
 - fundamental and classical supervised learning methods,
 - Bayesian statistics,
 - regularization,
 - nonparametric regression.
- **Second Half:**
 - neural networks,
 - Bayesian computation,
 - variational inference,
 - modern generative models and transformers.

First Half: Classical Learning and Bayesian Foundations

- Bayesian philosophy and probabilistic modelling
- Linear Regression
- Logistic Regression and Generalized Linear Models
- Frequentist and Bayesian perspectives on regression
- Discriminant Analysis for Classification: LDA and QDA
- Cross Validation
- Bootstrap
- Gaussian Sequence Model (Bayes and Empirical Bayes)
- James–Stein Estimator as Empirical Bayes
- Dirichlet–Multinomial Inference
- Ridge Regression from a Bayesian perspective
- Lasso and some of its properties

First Half: Nonlinear Curve Fitting

- We will also study nonlinear curve fitting methods.
- Topics include:
 - Smoothing Splines, Local Polynomial Regression
 - Gaussian Process Regression
 - Trend Filtering
 - Isotonic Regression
- These methods show how flexible function estimation can be done beyond ordinary linear models.
- We will view these methods through modelling, computation, and inference.

Second Half: Neural Networks and Bayesian Computation

- Neural Networks
 - basic architecture,
 - activation functions,
 - loss functions,
 - backpropagation.
- Bayesian Computation
 - Metropolis–Hastings,
 - Gibbs Sampling (Application to Mixture Models and connection to EM algorithm),
 - Hamiltonian Monte Carlo.
- Variational Inference
 - ELBO maximization,
 - connection to the EM algorithm,
 - Variational Autoencoders.

Second Half: Modern Generative AI

- We will discuss some modern generative models.
- Topics include:
 - Variational Autoencoders,
 - Diffusion Denoising Probabilistic Models,
 - Transformers in Large Language Models.
- We will emphasize the probabilistic structure wherever possible.
- For example, diffusion models are naturally connected to ideas from variational inference.

Modern AI will be presented as part of a longer statistical story.

Part 4: How the Course Will Be Taught

Lecture Notes, Code, and Demonstrations

- I will write lecture notes for every class.
- I will teach from these notes.
- We will also see Python code and notebook demonstrations in class.
- We will fit models, run algorithms, visualize outputs, and examine examples.

What This Course Will Not Emphasize

- This course will not be centered around frequentist risk calculations.
- We will not focus heavily on:
 - minimax rates,
 - MSE bounds,
 - concentration inequalities,
 - asymptotic theory for every method.
- These are important topics, but they form a different kind of course.
- Our focus will be on probabilistic modelling, posterior distributions, sampling, computation, prediction, and uncertainty.

A Course We Will Explore Together

- This is the first time I am teaching a course of this form.
- I will also be exploring, learning, and relearning parts of the material along with you.
- The goal is not to memorize a laundry list of algorithms.
- The goal is to understand a coherent set of ideas that connects statistics, probability, computation, and modern machine learning.

Modern AI is not a disconnected collection of algorithms.

Main Takeaway

This course will study machine learning through the lens of probabilistic inference.

Bayesian thinking will provide a unifying principle.

Computation will tell us how far we can actually carry out that principle.